

Plan

- Determinant
- Laplace Expansion Theorem
- Determinants of Elementary Matrices
- Cramer's Rule

Determinant

Definition

Let $A = [a_{ij}]$ be an $n \times n$ matrix.

- If $A = [a]$, **determinant** of A is defined as $\det(A) = a$.
- If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then we define $\det(A) = ad - bc$.
- Let A_{ij} be the submatrix of A obtained by deleting the i -th row and the j -th column of A .
- In general, $\det(A)$ is defined recursively as follows:

$$\begin{aligned}\det(A) &= a_{11} \det(A_{11}) - a_{12} \det(A_{12}) + \cdots + (-1)^{1+n} a_{1n} \det(A_{1n}) \\ &= \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A_{ij}\end{aligned}$$

- Sometimes $\det(A)$ is also denoted by $|A|$.
- We define $\det(A_{ij})$ to be the (i, j) -minor of A .
- The number $C_{ij} = (-1)^{i+j} \det(A_{ij})$ is called the (i, j) -cofactor of A .
- Thus we can write

$$\det(A) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}) = \sum_{j=1}^n a_{1j} C_{1j}.$$

Result (Properties of Determinants)

Let $A = [a_{ij}]$ be an $n \times n$ matrix. Then

$$\begin{aligned}\det(A) &= a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n} = \sum_{j=1}^n a_{1j}C_{1j} \\ &= a_{11}C_{11} + a_{21}C_{21} + \dots + a_{n1}C_{n1} = \sum_{i=1}^n a_{i1}C_{i1}\end{aligned}$$

Proof. By induction on n . True for $n = 1$. Assume true for all matrices of order $(n - 1) \times (n - 1)$. Now, let $A = [a_{ij}]$ be $n \times n$. We show that

$$a_{12}C_{12} + \dots + a_{1n}C_{1n} = a_{21}C_{21} + \dots + a_{n1}C_{n1}.$$

For $2 \leq i, j \leq n$, let us compute C_{1j} and C_{i1} .

$$\begin{aligned}
C_{1j} &= (-1)^{1+j} \det \begin{bmatrix} a_{21} & \cdots & a_{2,j-1} & a_{2,j+1} & \cdots & a_{2,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{i1} & \cdots & a_{i,j-1} & a_{i,j+1} & \cdots & a_{i,n} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{n,j-1} & a_{n,j+1} & \cdots & a_{nn} \end{bmatrix} \\
&= (-1)^{1+j} \sum_{i=2}^n (-1)^{(i-1)+1} a_{i1} \det A_{1i,1j} \text{ (using induction hypo.)} \\
&= \sum_{i=2}^n (-1)^{i+j+1} a_{i1} \det A_{1i,1j}
\end{aligned}$$

Therefore,

$$\sum_{j=2}^n a_{1j} C_{1j} = \sum_{j=2}^n \sum_{i=2}^n (-1)^{i+j+1} a_{i1} a_{1j} \det A_{1i,1j}$$

$$\begin{aligned}
C_{i1} &= (-1)^{i+1} \det \begin{bmatrix} a_{12} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{i-1,2} & \cdots & a_{i-1,j} & \cdots & a_{i-1,n} \\ a_{i+1,2} & \cdots & a_{i+1,j} & \cdots & a_{i+1,n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ a_{n2} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix} \\
&= (-1)^{i+1} \sum_{j=2}^n (-1)^{1+(j-1)} a_{1j} \det A_{1i,1j} \\
&= \sum_{j=2}^n (-1)^{i+j+1} a_{1j} \det A_{1i,1j}
\end{aligned}$$

Therefore,

$$\sum_{i=2}^n a_{i1} C_{i1} = \sum_{i=2}^n \sum_{j=2}^n (-1)^{i+j+1} a_{i1} a_{1j} \det A_{1i,1j} = \sum_{j=2}^n a_{1j} C_{1j}.$$

Result (Properties of Determinants)

Let A be an $n \times n$ matrix. If B is obtained by interchanging any two rows of A , then $\det(B) = -\det(A)$.

Proof. By induction on n . True for $n = 2$. Assume true for all matrices of order $(n - 1) \times (n - 1)$. Let $A = [a_{ij}]$, $B = [b_{ij}]$ be

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{r1} & \cdots & a_{rj} & \cdots & a_{rn} \\ a_{r+1,1} & \cdots & a_{r+1,j} & \cdots & a_{r+1,n} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}, \quad B = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{r+1,1} & \cdots & a_{r+1,j} & \cdots & a_{r+1,n} \\ a_{r1} & \cdots & a_{rj} & \cdots & a_{rn} \\ \vdots & & \vdots & & \vdots \\ a_{n1} & \cdots & a_{nj} & \cdots & a_{nn} \end{bmatrix}$$

Let $\det A = \sum_{i=1}^n a_{i1} C_{i1}$, $\det B = \sum_{i=1}^n b_{i1} C'_{i1}$. For $i \notin \{r, r+1\}$

$$b_{i1} C'_{i1} = a_{i1} (-1)^{i+1} \det B_{i,1} = -a_{i1} (-1)^{i+1} \det A_{i,1} = -a_{i1} C_{i1} \text{ (why?)}$$

$$\begin{aligned} b_{r1} C'_{r1} &= (-1)^{r+1} a_{r+1,1} \det B_{r1} = -a_{r+1,1} (-1)^{(r+1)+1} \det A_{r+1,1} \\ &= -a_{r+1,1} C_{r+1,1}. \end{aligned}$$

$$b_{r+1,1} C'_{r+1,1} = (-1)^{r+2} a_{r1} \det B_{r+1,1} = -a_{r1} (-1)^{r+1} \det A_{r,1} = -a_{r1} C_{r1}.$$

Therefore,

$$\det B = \sum_{i=1}^n b_{i1} C'_{i1} = \sum_{i=1}^n -a_{i1} C_{i1} = -\det A.$$

- We have proved that if B is obtained from A by interchanging two adjacent rows, then $\det B = -\det A$.
- If $s > r$ and B is obtained from A by interchanging the r -th and the s -th rows of A , then this can be done with $2k - 1$ successive interchanges of adjacent rows, where $k = s - r$. Hence,

$$\det B = (-1)^{2k-1} \det A = -\det A.$$

Result (Laplace Expansion Theorem)

The determinant of an $n \times n$ matrix $A = [a_{ij}]$, where $n \geq 2$, can be computed as

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in} = \sum_{j=1}^n a_{ij}C_{ij},$$

(this is the *cofactor expansion along the i -th row*),
and also as

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj} = \sum_{i=1}^n a_{ij}C_{ij},$$

(this is the *cofactor expansion along the j -th column*).

Result

- For any square matrix A , $\det(A^t) = \det(A)$.
- If $A = [a_{ij}]$ is an $n \times n$ *triangular* matrix then $\det(A) = a_{11}a_{22} \dots a_{nn}$.

Result (Properties of Determinants)

- If A has a *zero row* then $\det(A) = 0$.
- If A has two *identical rows* then $\det(A) = 0$.
- If B is obtained by *multiplying a row* of A by k , then $\det(B) = k\det(A)$.
- If the matrices A , B and C are identical except that the i -th row of C is the sum of the i -th rows of A and B , then $\det(C) = \det(A) + \det(B)$.
- If C is obtained by adding a *multiple of one row* of A to another row, then $\det(C) = \det(A)$.
- If $A = [a_{ij}]$, where $n \geq 2$, then $a_{i1}C_{j1} + a_{i2}C_{j2} + \dots + a_{in}C_{jn} = 0$ for $i \neq j$.

Result (**Determinants of Elementary Matrices**)

Let E be an $n \times n$ elementary matrix. Then

- 1 $\det(E) = -1, k$ or 1 (for types 1, 2 and 3, respectively).
- 2 For any $n \times n$ matrix A , $\det(EA) = \det(E) \det(A)$.

Let A be $n \times n$ and $R = \text{RREF}(A)$. Then $A = E_k \cdots E_1 R$ for some elementary matrices E_i . We have

$$\det(A) = \det(E_k) \cdots \det(E_1) \det(R).$$

Result

- A square matrix A is *invertible* if and only if $\det(A) \neq 0$.
- Let A be an $n \times n$ matrix. Then $\det(kA) = k^n \det(A)$.
- Let A and B be two $n \times n$ matrices. Then

$$\det(AB) = \det(A) \det(B).$$

- If the matrix A is invertible then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Result

Let the matrices $\begin{bmatrix} P & R \\ \mathbf{0} & Q \end{bmatrix}$, P and Q be square matrices.

Then $\det \begin{bmatrix} P & R \\ \mathbf{0} & Q \end{bmatrix} = \det(P) \det(Q)$.

★ A matrix A is said to be **singular** or **non-singular** according as $\det(A) = 0$ or $\det(A) \neq 0$.

★ A determinant can be expanded **column-wise**.

★ All the previous results based on row-wise expansion of determinant are also **valid** for **column-wise** expansion.

Result

- If B is obtained by interchanging any two *columns* of A , then $\det(B) = -\det(A)$.
- If A has a *zero column* then $\det(A) = 0$.
- If A has two *identical columns* then $\det(A) = 0$.
- If B is obtained by *multiplying a column* of A by k , then $\det(B) = k \det(A)$.
- If A , B and C are identical except that the i -th column of C is the sum of the i -th columns of A and B , then $\det(C) = \det(A) + \det(B)$.
- If C is obtained by adding a *multiple of one column* of A to another column, then $\det(C) = \det(A)$.
- If $A = [a_{ij}]$, where $n \geq 2$, then $a_{1i}C_{1j} + a_{2i}C_{2j} + \dots + a_{ni}C_{nj} = 0$ for $i \neq j$.

Definition

Let A be an $n \times n$ matrix and let $\mathbf{b} \in \mathbb{R}^n$. Then $A_i(\mathbf{b})$ denotes the matrix obtained by replacing the i -th column of A by $\mathbf{b}$. That is, if $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n]$, then $A_i(\mathbf{b}) = [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_{i-1} \ \mathbf{b} \ \mathbf{a}_{i+1} \ \dots \ \mathbf{a}_n]$.

Result (Cramer's Rule)

Let A be an $n \times n$ invertible matrix and let $\mathbf{b} \in \mathbb{R}^n$. Then the *unique* solution $\mathbf{x} = [x_1, x_2, \dots, x_n]^t$ of the system $A\mathbf{x} = \mathbf{b}$ is given by

$$x_i = \frac{\det(A_i(\mathbf{b}))}{\det(A)} \quad \text{for } i = 1, 2, \dots, n.$$

The Adjoint of a Matrix: Let $A = [a_{ij}]$ be an $n \times n$ matrix and let C_{ij} be the (i, j) -cofactor of A . Then the **adjoint** of A , denoted $\text{adj}(A)$, is defined as

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \vdots & \vdots & \vdots & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix} = [C_{ij}]^t.$$

Result

Let A be an $n \times n$ invertible matrix. Then $A^{-1} = \frac{1}{\det(A)} \cdot \text{adj}(A)$.

Exercise

Use the **adjoint method** to find the **inverse** of the matrix

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 4 \\ 1 & 3 & -3 \end{bmatrix}.$$